

LYES DJEMAA

Software Engineering Internship

Distributed Systems · Backend · Blockchain

Available from mid-February 2027 · 4-6 months

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Final-year Master's student in Computer Science with a dual Bachelor's degree in Mathematics and Computer Science. Built projects in distributed systems, blockchain and parallel computing, with a focus on reliable software.

EDUCATION

Université Sorbonne Paris Nord, France

Master's in Computer Science - Final year (M2)

2026 - 2027

P2S track: Programming, Software Safety and Security.

Coursework: software engineering, program specification and verification.

Master's in Computer Science - First year (M1)

2025 - 2026

Dual Bachelor's degree in Mathematics and Computer Science

2022 - 2025

TECHNICAL SKILLS

Languages Python, Java, C, Rust, C++, SQL.

Software REST APIs, Flask, SQLAlchemy; unit testing, debugging, documentation.

Systems MPI, Java RMI, sockets, threads; process synchronization and coordination.

Tools Git/GitHub, Docker, Linux, Bash, Cargo, CMake, pytest.

PERSONAL AND ACADEMIC PROJECTS

QChain Blockchain

Python · Flask · Docker

- Developed a Proof-of-Work blockchain with transactions, wallets, a mempool, node synchronization and fork selection based on cumulative work.
- Built a REST API and CLI; set up a **three-node Docker testnet** and automated tests.

Rust Matching Engine

Rust · Cargo

- Implemented order placement and cancellation, partial fills, and best bid/ask tracking; verified matching logic with unit tests.

Distributed Resource Manager

Java · RMI · Sockets

- Designed a two-phase commit prototype (prepare/commit/abort) with a coordinator, participants and locks for each resource.

Parallel Gauss-Jordan Elimination

C · MPI

- Parallelized Gauss-Jordan elimination across MPI processes and coordinated interprocess data exchanges.

ADDITIONAL EXPERIENCE

Guest Services Host | Mahola

May - August 2026

Welcomed and directed visitors, working with a team to provide clear, professional assistance.

LANGUAGES AND STRENGTHS

French: Native · **English:** Professional proficiency · **TOEIC 900/990**

Analytical thinking, attention to detail, independence and adaptability.